Linear Algebra

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Chapter 1

Linear Systems

1.1 Solving Linear Systems

1.1.1 Gauss's Method

Definition 1.1.1 (Linear combinations)**.** A *linear combination* of $x_1, x_2, ..., x_n$ has the form

$$a_1x_1 + a_2x_2 + ... + a_nx_n$$

where the numbers $a_1, ..., a_n \in \mathbb{R}$ are the combination's *coefficients*. A *linear equation* in the variables $x_1, ..., x_n$ has the form $a_1x_1 + a_2x_2 + ... + a_nx_n = d$ where $d \in \mathbb{R}$ is the *constant*.

An *n-tuple* $(s_1, s_2, ..., s_n) \in \mathbb{R}^n$ is a *solution* of, or *satisfies*, that equation if substituting the numbers $s_1, ..., s_n$ for the variables gives a true statement: $a_1s_1 + a_2s_2 + ... + a_ns_n = d$. A *system of linear equation*

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= d_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= d_2 \\ \vdots \qquad\qquad\qquad\qquad &\quad \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= d_m \end{aligned}$$

has the *solution* $(s_1, s_2, ..., s_n)$ if that *n-tuple* is a solution of all of the equations.

Find the set of all solutions is *solving* the system.

Definition 1.1.2 (Elementary operations)**.** If a linear system is changed to another one by one of these operations

- an equation is swapped with another
- an equation has both sides multiplied by a nonzero constant
- an equation is replaced by the sum of itself and a multiple of another

then the two systems have the same set of solutions. These operations are known as the **elementary operations**.

Definition 1.1.3 (Leading variable)**.** In each row of a system, the first variable with a nonzero coefficient is the row's **leading variable**. A system is in **echelon form** if each leading variable is to the right of the leading variable in the row above it, except for the leading variable in the first row, and any rows with all-zero coefficients are at the bottom.

1.1.2 Exercises

1.17 a) $\begin{aligned} 2x+3y &= 13 \\ x-y &= -1 \end{aligned} \rightarrow \begin{aligned} x-y &= -1 \\ 5y &= 15 \end{aligned} \rightarrow \begin{aligned} x-y &= -1 \\ y &= 3 \end{aligned}, \quad \begin{aligned} x &= 2 \\ y &= 3 \end{aligned}$

b) $\begin{aligned} x \quad\quad -z &= 0 \\ 3x+y \quad\quad &= 1 \\ -x+y+z &= 4 \end{aligned} \rightarrow \begin{aligned} x \quad\quad -z &= 0 \\ y \quad\quad &= 4 \\ y+3z &= 1 \end{aligned} \rightarrow \begin{aligned} x \quad\quad -z &= 0 \\ y \quad\quad &= 4 \\ 3z &= -3 \end{aligned} \rightarrow \begin{aligned} x &= -1 \\ y &= 4 \\ z &= -1 \end{aligned}$

1.19 a) $\begin{aligned} 2x+2y &= 5 \\ x-y &= 0 \end{aligned} \rightarrow \left[\begin{array}{cc|c} 2 & 2 & 5 \\ 1 & -4 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 5/2 \\ 0 & -5 & -5/2 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 5/2 \\ 0 & 1 & 1/2 \end{array}\right]$, Unique Solution

b) $\begin{aligned} -x+y &= 1 \\ x+y &= 2 \end{aligned} \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ -1 & 1 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 2 & 3 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 3/2 \end{array}\right]$, Unique Solution

c) $\begin{aligned} x-3y+z &= 1 \\ x+y+2z &= 14 \end{aligned} \rightarrow \left[\begin{array}{ccc|c} 1 & -3 & 1 & 1 \\ 1 & 1 & 2 & 14 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -3 & 1 & 1 \\ 0 & -2 & 3 & 15 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -3 & 1 & 1 \\ 0 & 1 & -3/2 & -15/2 \end{array}\right]$, Infinite Solutions

d) $\begin{aligned} x-y &= 1 \\ -3x-3y &= 2 \end{aligned} \rightarrow \left[\begin{array}{cc|c} -1 & -1 & 1 \\ -3 & -3 & 2 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 5 \end{array}\right]$, Infinite Solutions

e) $\begin{aligned} 4y+z &= 20 \\ 2x-2y+z &= 0 \\ x \quad\quad +z &= 5 \\ x+y-z &= 10 \end{aligned} \rightarrow \left[\begin{array}{ccc|c} 0 & 4 & 1 & 20 \\ 2 & -2 & 1 & 0 \\ 1 & 0 & 1 & 5 \\ 1 & 1 & -1 & 10 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 10 \\ 0 & 4 & 1 & 20 \\ 0 & -4 & 3 & -20 \\ 0 & -1 & 2 & -5 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 10 \\ 0 & 1 & 1/4 & 5 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 9/4 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 10 \\ 0 & 1 & 1/4 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$, Unique Solution

f) $\begin{aligned} 2x \quad\quad +z+w &= 5 \\ y \quad\quad -w &= -1 \\ 3x \quad -z-w &= 0 \\ 4x+y+2z+w &= 9 \end{aligned} \rightarrow \left[\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 5 \\ 0 & 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 & 0 \\ 4 & 1 & 2 & 1 & 9 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1/2 & 1/2 & 5/2 \\ 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 1 & 0 & -1 & -1 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1/2 & 1/2 & 5/2 \\ 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$, Infinite Solutions

1.21 a) $x = 1-3y$; $2x+y = 2(1-3y)+y = 2-6y+y = -5y+2 = -3$, $y = \frac{-3-2}{-5} = 1$

b) $x = 1-3y$; $2x+2y = 2(1-3y)+2y = 2-6y+2y = -4y+2 = 0$, $y = \frac{-2}{-4} = 1/2$

Solve all equations in the system to find consistency

1.22 $\begin{aligned} x-y &= 1 \\ 3x-3y &= k \end{aligned} \rightarrow \begin{aligned} x-y &= 1 \\ 0 &= k-3 \end{aligned}$, No solutions for $k \neq 3$
Infinite solutions for $k = 3$
System can't have a unique solution for any k

1.24 a)

$$\begin{aligned} x-3y&=b_1\\ 3x+y&=b_2\\ x+7y&=b_3\\ 2x+4y&=b_4 \end{aligned} \rightarrow \begin{aligned} x-3y&=b_1\\ 10y&=-3b_1+b_2\\ 10y&=-b_1+b_3\\ 10y&=-2b_1+b_4 \end{aligned} \rightarrow \begin{aligned} x-3y&=b_1\\ 10y&=-3b_1+b_2\\ 0&=2b_1-b_2+b_3\\ 0&=b_1-b_2+b_4 \end{aligned}, \quad \begin{aligned} 2b_1-b_2+b_3&=0\\ b_1-b_2+b_4&=0 \end{aligned}$$

1.27 $f(x)=ax^2+bx+c$, $f(1)=2$, $f(-1)=6$, $f(2)=3$

$$\begin{aligned} f(1)&=a+b+c=2\\ f(-1)&=a-b+c=6\\ f(2)&=4a+2b+c=3 \end{aligned}, \quad \begin{aligned} a+b+c&=2\\ -2b&=4\\ -2b-3c&=-5 \end{aligned} \rightarrow \begin{aligned} a+b+c&=2\\ b&=-2\\ -3c&=-9 \end{aligned} \rightarrow \begin{aligned} a+b+c&=2\\ b&=-2\\ c&=3 \end{aligned} \rightarrow \begin{aligned} a&=2-b-c=2+2-3=1\\ b&=-2\\ c&=3 \end{aligned}$$

$f(x)=x^2-2x+3$

1.32 Unique solution: Lines are not parallel, they intersect at a single point
No solution: Lines are parallel and not equivalent, they never intersect
Infinite Solutions: Lines are parallel and equivalent

1.35 Swaping is redundant, $r_a \leftrightarrow r_b$ can be replaced by $r_a+r_b \rightarrow r_a$, $-(r_b-r_a') \rightarrow r_b$, $r_a'-r_b' \rightarrow r_a$

1.37

$$\begin{aligned} ap+bn+cd&=0.83\\ a+b+c&=13 \end{aligned} \rightarrow \begin{aligned} 0.01a+0.05b+0.10c&=0.83\\ a+b+c&=13 \end{aligned} \rightarrow \begin{aligned} a+5b+10c&=83\\ a+b+c&=13 \end{aligned} \rightarrow \begin{aligned} a+5b+10c&=83\\ b+\tfrac{9}{4}c&=70/4 \end{aligned}, \quad (a,b,c)=(3,4,6)$$

1.38

$$\begin{aligned} a+\tfrac{1}{3}b+\tfrac{1}{3}c+\tfrac{1}{3}d&=23\\ \tfrac{1}{3}a+b+\tfrac{1}{3}c+\tfrac{1}{3}d&=21\\ \tfrac{1}{3}a+\tfrac{1}{3}b+c+\tfrac{1}{3}d&=17\\ \tfrac{1}{3}a+\tfrac{1}{3}b+\tfrac{1}{3}c+d&=29 \end{aligned} \rightarrow \begin{aligned} a&=12\\ b&=9\\ c&=3\\ d&=21 \end{aligned}, \quad \underline{(b)\ 21}$$

1.39

$$\begin{aligned} T&=A+1\\ +10+E&=H+T\\ H&=A+E+1\\ +10+A&=H+H\\ W&=A+E \end{aligned} \rightarrow \begin{array}{ccccc} A & E & H & T & W\\ A & & & -T & =-1\\ & -E & +H & +T & =10\\ A & +E & -H & & =-1\\ -A & & 2H & & =10\\ -A & -E & & & W=0 \end{array} \rightarrow \begin{aligned} A&=4\\ E&=2\\ H&=7\\ T&=5\\ W&=6 \end{aligned}$$

1.40

$$\underset{a}{A>B}\ (11),\ \underset{b}{B>A}\ (9),\ \underset{c}{C>A}\ (12),\ \underset{d}{A>C}\ (8),\ \underset{e}{B>C}\ (14),\ \underset{f}{C>B}\ (6)$$

$$\underset{a}{ABC},\ \underset{b}{ACB},\ \underset{c}{BAC},\ \underset{d}{BCA},\ \underset{e}{CAB},\ \underset{f}{CBA}$$

8 people voted B first

$a=6,\ b=1,\ c=1,\ d=7,\ e=4,\ f=1$

$$\left[\begin{aligned} a+b+c+d+e+f&=20\\ a+b\quad +e&=11\\ d+e+f&=12\\ a\quad +c+d&=14\\ +c+d\quad +f&=9\\ a+b+c&=8\\ b\quad +e+f&=6 \end{aligned}\right] \rightarrow \left[\begin{aligned} a+b+c+d+e+f&=20\\ a+b\quad +e&=11\\ d+e+f&=12\\ a\quad +c+d&=14\\ b+c\quad +f&=3 \end{aligned}\right]$$

1.41

$$\begin{aligned} ax + (a+d)y &= (a+2d) \\ (a+3d)x + (a+4d)y &= (a+5d) \end{aligned} \rightarrow \begin{aligned} ax + (a+d)y &= (a+2d) \\ 3d^2 y &= 6d^2 \end{aligned} \rightarrow \begin{aligned} x &= -1 \\ y &= 2 \end{aligned}$$

1.2 Linear Geometry

1.3 Reduced Echelon Form

Chapter 2

Vector Spaces

Chapter 3

Maps Between Spaces

Chapter 4

Determinants

Chapter 5

Similarity